

Study Questions: Action

1. How does Active Inference reinterpret motor commands as proprioceptive predictions rather than explicit commands to muscles?
2. Explain the role of the spinal reflex arc in the Active Inference account of motor control.
3. How does this framework reinterpret the stretch reflex — is it a primitive response or the final stage of hierarchical inference?
4. What are saccadic eye movements, and how does Expected Free Energy explain their patterns?
5. Distinguish epistemic action from pragmatic action. Give three examples of each.
6. How does haptic exploration (touching and feeling objects) serve as epistemic action?
7. Explain how loss of dopaminergic neurons in Parkinson's disease leads to bradykinesia through reduced motor precision.
8. What is a functional motor disorder, and how does Active Inference explain symptoms with no identifiable neurological cause?
9. How does the concept of a "prior belief about inability" create real motor paralysis in functional disorders?
10. What is apraxia, and how does it illustrate the hierarchical nature of the motor model?
11. Compare the Active Inference account of motor control with the classical optimal control theory (Todorov & Jordan, 2002).
12. How does the cerebellum contribute to motor prediction and prediction error computation?
13. What role does the basal ganglia play in precision-weighting motor policies?
14. How does Active Inference explain the phenomenon of motor imagery (imagining an action without performing it)?
15. What is the ideomotor principle (James, 1890), and how does Active Inference formalize it?
16. How does Active Inference account for the speed-accuracy trade-off in motor control?
17. What is the role of the supplementary motor area (SMA) in sequential action planning?
18. How does mirror neuron activity relate to motor prediction — observing another's actions as "predicting" their motor trajectory?

19. Explain how deep brain stimulation for Parkinson's disease might work through precision modulation.
20. Design an experiment to test whether saccadic eye movements to a naturalistic scene follow the Expected Free Energy predictions.